Hanyi Zhao

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Research Interests

My research focuses on robotic visual perception and planning in dynamic, unstructured environments, aiming to enable robots to understand, adapt, and manipulate effectively across diverse real-world scenarios.

Education

- **Harbin Institute of Technology, Shenzhen** *Sep 2022 - present*
  - Degree: B.E. in Automation; GPA: 91.758/100
  - Awards: National Scholarship, First-Class Academic Scholarship (two times)

Publications

[1] **H. Zhao***, J. Zhu*, Z. Yan*, Y. Li, Y. Deng†, and X. Wang†, "Learning generalizable language-conditioned cloth manipulation from long demonstrations", *International Conference on Intelligent Robots and Systems (**IROS**)*, 2025.[Paper] [Website] [Code]

Experience

- **Center for Artificial Intelligence and Robotics, Tsinghua SIGS** *Oct 2023 - Feb 2025*
  Research Intern with Prof.Xueqian Wang *Shenzhen,China*
  - Developed a novel pipeline that extracts reusable skills from existing long demonstrations and composes them to generalize across unseen multi-step cloth manipulation tasks.
  - Introduced an autonomous skill discovery framework for cloth manipulation, leveraging an LLM to provide commonsense knowledge for task decomposition.
  - Conducted extensive experiments in both simulation and real-world settings, validating the effectiveness and generalization of the proposed approach.
- **Adaptive Computing Lab & Smart Systems Institute, NUS** *Feb 2025 - Present*
  Research Intern with Prof.David Hsu *Singapore*
  - Designed a real-time 3D scene graph construction framework that integrates each object's mesh, pose, and inter-object relations, while simultaneously reconstructing the background geometry.
  - Introduced a tracking module that fuses end-effector motion with observations to accurately estimate and update object poses during manipulation.
  - Utilized spatial relationships within the scene graph to preserve the consistency and continuity of moving objects through temporary invisibility or occlusion.

Projects

- **RoboMaster Aerial Robot Design and Optimization** *Nov 2022 - Aug 2023*
  RoboMaster Competition 2023
  - Designed and built a lightweight, high-performance UAV featuring a shooting gimbal, an airframe, propeller guards, and other custom structural components.
  - Constructed the full system using advanced prototyping techniques such as 3D printing, laser cutting, and precision machining.
- **Autonomous Mobile Robot Design and Development** *Dec 2023 - Jul 2024*
  The 19th National University Students Intelligent Car Race
  - Designed and implemented an autonomous mobile robot, capable of efficiently performing diverse tasks such as trajectory following, target detection, and visual classification.
  - Developed the complete mechanical, control, and perception system, including motion control, serial communication, and image processing algorithms, enabling adaptive and accurate task execution in complex environments.

Skills

- **Programming Languages:** Python, C++, C, Matlab
- **Softwares & Tools:** Linux, ROS, PyTorch, OpenCV, Solidworks, Latex, Windows
- **Hardware:** Multiple Sensors, OptiTrack Motion Capture System, STM32, Basic Mechanical Design and Assembly
- **Languages:** English (fluent: IELTS 7.5), Chinese (native)

Awards & Certifications

- H Award of the Mathematical Contest in Modeling(MCM) *May 2024*
- First Prize in Guangdong Province of the Blue Bridge Cup National Software and Information Technology Professional Talent Competition *Jun 2024*
- Second Prize in the South China Division of the 19th National University Students Intelligent Car Race *Jul 2024*